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Lithium-Ion Battery Simulations with Complex 3D Porous Structures

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Abstract. This study presents an advanced simulation framework for lithium-ion batteries, developed under the project "CODEVELOP-GT/0322/0081 - REAL Live SimULATION of GREEN software in the CLOUD," a collaboration between KYAMOS LTD and Frederick University. The methodology utilizes the Lattice Boltzmann Method with a Multiple Relaxation Time collision model and a D3Q27 lattice Boltzmann stencil in three-dimensional space to model complex bicontinuous porous structures for the anode and cathode. This approach accurately simulates critical electrochemical and transport processes, including mass transport, charge transfer, and fluid dynamics, within intricate electrode microstructures. The LBM-MRT framework ensures numerical stability and precision, supported by a solver written in C++ CUDA that harnesses Nvidia GPUs for efficient parallel processing. The solver is paired with an intuitive GUI, allowing users to define boundary and initial conditions, create unique bicontinuous microstructures, and visualize results such as lithium-ion concentration, electric field values, and current/voltage profiles. This comprehensive tool facilitates the design and optimization of lithium-ion batteries, offering valuable insights into next-generation energy storage solutions. Conducted within the CODEVELOP-GT initiative, this work integrates cloud-based green software principles with high-performance computing, contributing to sustainable technology development. The collaboration between KYAMOS LTD and Frederick University unites industry innovation with academic expertise, delivering a scalable and user-friendly platform for advanced battery simulations that pushes the boundaries of computational electrochemistry.

1. Introduction

Lithium-ion batteries (LIBs) have become the cornerstone technology for modern portable electronics, electric vehicles (EVs), and grid energy storage due to their high energy density, long cycle life, and continuous advancements in electrode and electrolyte materials. Beyond their dominant role in mobility, LIBs are increasingly deployed as stationary energy storage systems integrated with photovoltaic (PV) installations. Coupling PV generation with LIB banks allows surplus solar energy to be stored during peak production and released when sunlight is unavailable, improving energy reliability and enabling greater renewable penetration [1].



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The performance and reliability of LIBs are intricately linked to phenomena occurring across multiple spatial and temporal scales, ranging from atomistic interactions and particle-level processes within electrode microstructures to cell-level thermal and mechanical effects. To thoroughly understand and optimize LIB behaviour, advanced modelling and simulation approaches are essential [2].

Over the past decades, various modelling strategies have been developed, including empirical, equivalent-circuit, and physics-based models. Mechanism-based modelling frameworks—such as the pseudo-two-dimensional (P2D) model—capture electrolyte and charge transport, phase transformations, and interfacial reactions by solving coupled partial differential equations [2]. However, these models often rely on homogenized representations of electrode microstructures, potentially missing critical aspects of tortuosity, porosity, and local transport phenomena that are essential for next-generation batteries with complex architectures [2].

Consequently, increasing attention has been given to modelling techniques that directly represent three-dimensional microstructural characteristics. The Lattice Boltzmann Method (LBM) has emerged as a particularly powerful tool for simulating multiphysics transport processes in complex porous media, including those found in LIB electrodes [3]. Notably, three-dimensional (3D) pore-scale LBM models have enabled detailed simulations of ion, electron, and fluid transport—alongside electrochemical reactions—within realistic electrode geometries reconstructed from imaging or generated stochastically. Such approaches have illuminated how microstructural descriptors (e.g., pore size, surface area, tortuosity) impact local lithium concentration, current densities, and rate performance, thereby guiding the rational design of advanced electrode architectures [3].

Furthermore, recent efforts focus on the development of user-friendly cloud-based platforms that support scalable, energy-efficient simulations while embracing sustainable software practices. Such integrated environments bridge the gap between detailed computational models and practical battery design workflows, unifying innovations from academic research and industrial development to accelerate the creation of high-performance, sustainable energy storage solutions [4]. By advancing microstructure-resolved, GPU-accelerated simulation tools and integrating cloud-based green software concepts, modern computational electrochemistry is poised to provide deeper insights into lithium-ion transport dynamics and optimize next-generation battery designs in a scalable and sustainable manner.

In the context of EV applications, charging time remains a critical performance parameter. Fast-charging technologies can restore approximately 80 % of an EV battery's capacity within 15–30 minutes under DC fast-charging conditions, depending on cell chemistry and temperature management [5]. Such rapid charging introduces steep electrochemical and thermal gradients, reinforcing the need for high-fidelity multiphysics models capable of resolving transient transport phenomena during high-power operation.

In this context, the present work advances LIB simulation by integrating a high-fidelity bi-continuous porous electrode generator with an LBM solver implemented on GPUs. The coupled solver models lithium-ion and electron transport and interfacial electrochemical reactions

governed by Butler–Volmer kinetics in fully 3D domains. A dedicated GUI facilitates intuitive domain creation, parameter input, and results visualization, bridging computational electrochemistry with practical battery design workflows.

2. Mathematical and Numerical Formulation

2.1 Electrochemical Model

The simulation captures coupled transport and reaction phenomena during LIB discharge at the pore scale. The governing macroscopic equations obtained from [6] are:

(i) lithium-ion transport in the solid active material, (ii) electron conduction in the solid phase

$$\frac{\partial c_s}{\partial t} - \nabla \cdot (D_s \nabla c_s) = 0 \quad (\text{i})$$

$$\nabla \cdot (\sigma \nabla \varphi_s) = 0 \quad (\text{ii})$$

(iii) lithium-ion transport in the electrolyte, and (iv) ionic potential in the electrolyte.

$$\frac{\partial C_s}{\partial t} - \nabla \cdot (D_e \nabla C_e) - \nabla \cdot \left(\frac{D_e z F C_e}{R T} \nabla \varphi_e \right) = 0 \quad (\text{iii})$$

$$\nabla \cdot (\kappa \nabla \varphi_e) - \nabla \cdot \left(\frac{D_e z F C_e}{R T} \nabla C_e \right) = 0 \quad (\text{iv})$$

with electrochemical reactions at the solid–electrolyte interface described by Butler–Volmer kinetics (v) :

$$I = i_0 \left[\exp \left(\frac{\alpha_a F \eta}{R T} \right) - \exp \left(\frac{-\alpha_c F \eta}{R T} \right) \right] \quad (\text{v})$$

where the exchange current density is given by the following equation (vi) :

$$i_0 = F k_r C_e^{\alpha_a} (C_{max} - C_{surf})^{\alpha_a} C_{surf}^{\alpha_c} \quad (\text{vi})$$

2.2 Bi-continuous Porous Structure Generation

An in-house algorithm generates 3D bi-continuous microstructures with tuneable porosity and tortuosity, based on [7]. The algorithm ensures full percolation of both solid and electrolyte phases, overcoming limitations of sphere-packing methods that yield disconnected or

oversimplified domains. This enables realistic transport pathways and interfacial areas that closely replicate actual electrode architectures.

2.3 Numerical Implementation

The LBM solver employs a multiple relaxation time (MRT) D3Q27 lattice for stability and accuracy. Streaming is phase-restricted to avoid nonphysical transport between solid and electrolyte phases. Flux boundary conditions couple the transport equations via Butler-Volmer reaction kinetics at interfaces. The solver is implemented in C++ with CUDA, exploiting GPU parallelism for computational efficiency.

2.4 Boundary and Initial Conditions

The domain is initialized with lithium-ion concentrations corresponding to a fully charged state, and electrode potentials are set to represent typical cell voltages (e.g., 4.2 V cathode, 0.1 V anode). Zero-gradient boundary conditions prevent artificial leakage of species and potentials on external domain faces, while galvanostatic current collector conditions are applied on the left and right domain boundaries.

2.5 Graphical User Interface

The GUI integrates OpenCascade for 3D domain visualization and bi-continuous structure creation, as can be seen in Figure 1. Users can define simulation parameters, assign boundary conditions, and launch simulations remotely. Results are visualized with PLplot, enabling detailed analysis of lithium-ion concentration, potential fields, and discharge profiles.

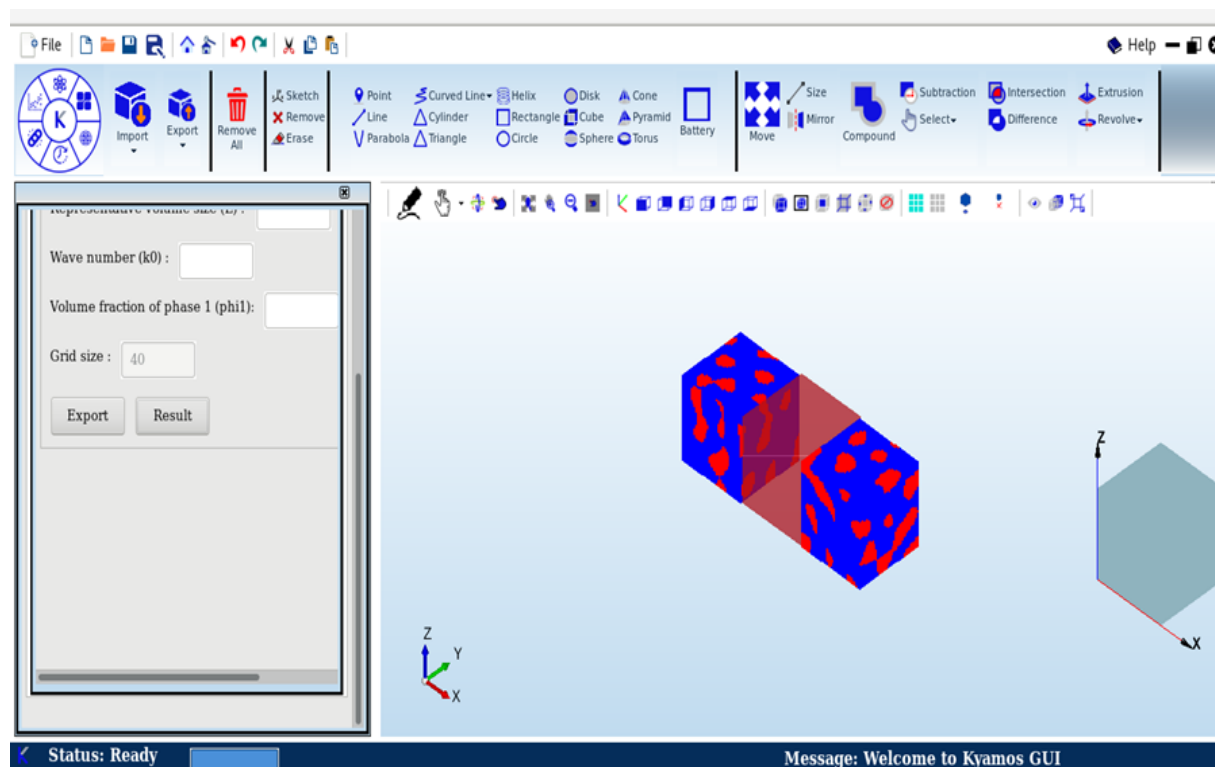


Figure 1. Generation and visualization of the complex, bi-continuous, 3D, porous electrode structures

3. Numerical experiments

Simulations shown in Figures 2,3,4 produce spatial and temporal maps of lithium-ion concentration and electrical potentials throughout the 3D domain comprising anode, separator, and cathode regions. Visualizations confirm physically consistent behavior: lithium deintercalates from the anode surface into the electrolyte and intercalates onto the cathode surface, reflecting the discharge process. The natural diffusion of lithium ions within the solid phases transports ions from bulk interiors to reactive surfaces.

Temporal evolution shows nearly complete depletion of lithium ions at the anode surface and corresponding accumulation at the cathode during discharge. Reverse reactions are suppressed by surface overpotentials as described by Butler-Volmer kinetics, consistent with thermodynamic and kinetic constraints. These results demonstrate that the boundary conditions and coupled LBM framework effectively capture the essential electrochemical physics.

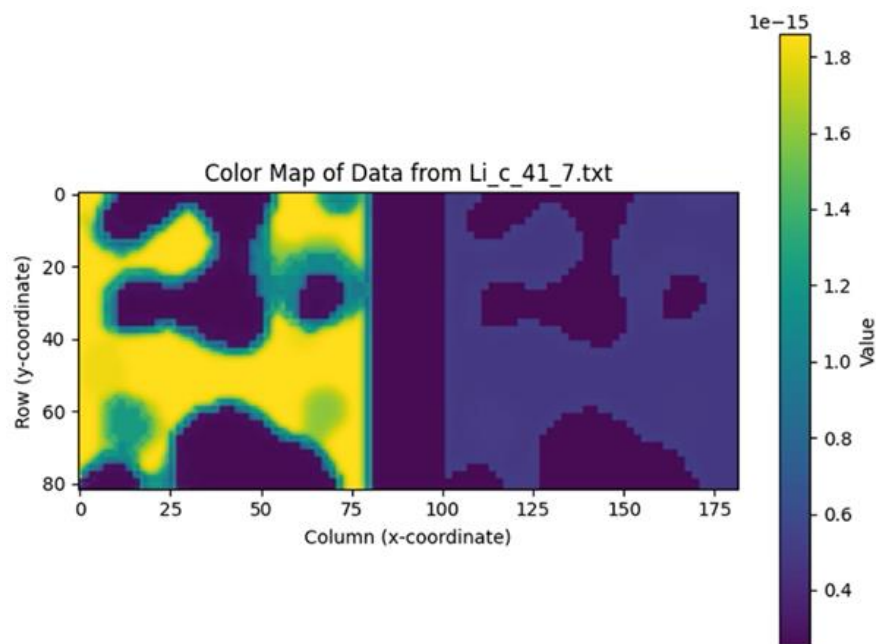


Figure 2: Initial Lithium-ion concentrations short after initialization. High concentration at the anode, low concentration at the cathode and almost zero concentration inside the electrolyte, as the battery is in a “charged” state.

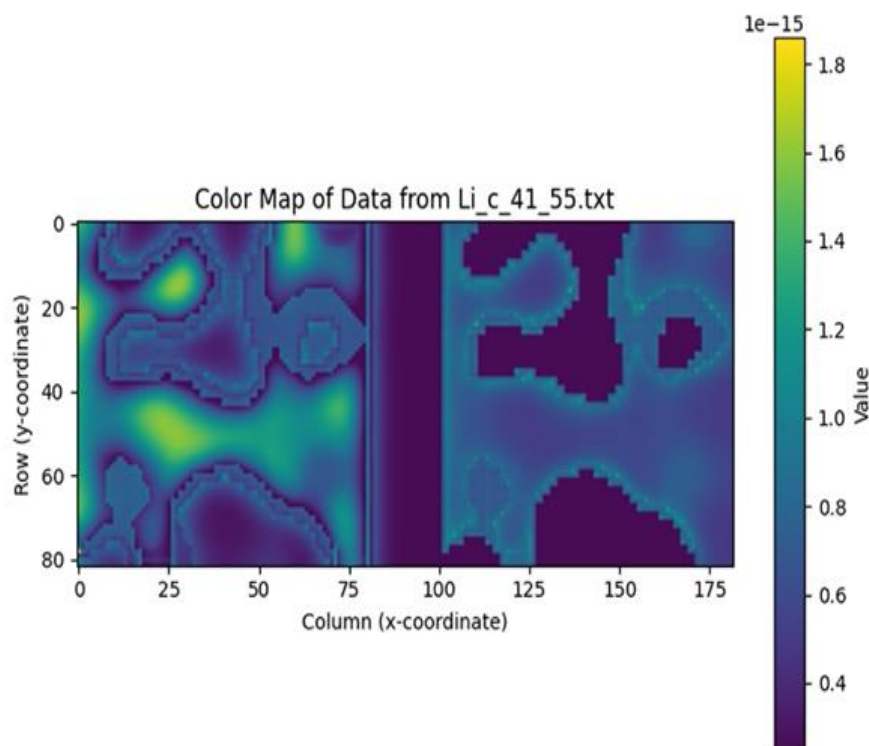


Figure 3: Anode (Left region), separator (center region) and cathode (right region) lithium concentrations as diffusion, deintercalation and intercalation take place

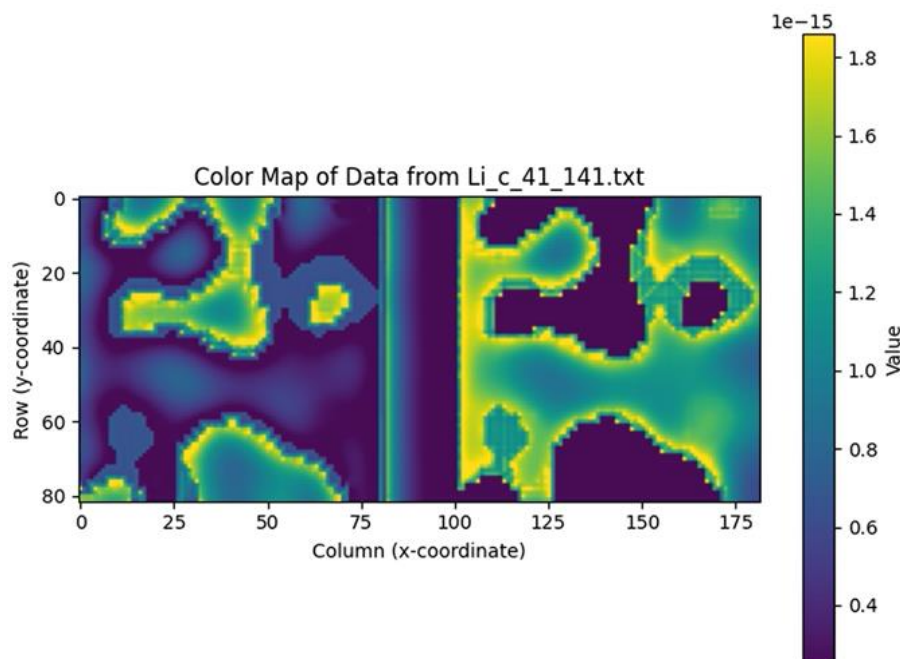


Figure 4. Deintercalation and intercalation processes at a later stage of the simulation. The cathode (right) shows significant lithium uptake from the surrounding electrolyte, while the anode exhibits minimal lithium insertion due to the interfacial overpotential governed by Butler–Volmer reaction kinetics.

4. Discussion

The presented framework advances lithium-ion battery modelling by resolving pore-scale electrochemical and transport processes within realistic 3D bicontinuous microstructures. Unlike traditional models that assume homogenized or simplified particle assemblies, our approach explicitly captures the complex connectivity of both solid active materials and electrolyte phases. This allows for more accurate prediction of tortuosity effects and interfacial reaction areas, which are critical for reliable performance estimation and optimization.

The use of a GPU-accelerated LBM solver enables high-resolution simulations within computationally feasible times, supporting parametric studies of microstructure-property relationships. The integration of a user-friendly GUI lowers the barrier for adoption by researchers and industry practitioners, bridging the gap between academic modelling advances and practical battery design workflows.

Our results validate the physical consistency of ion transport and interfacial kinetics under realistic boundary conditions. The observed lithium concentration gradients and reaction asymmetries align with expected discharge behaviours, confirming the accuracy of the Butler-Volmer coupling within the bi-continuous domain. This confirms the suitability of the framework for investigating degradation mechanisms linked to inhomogeneous lithium distribution, which can lead to capacity fade and performance loss.

Looking ahead, the framework can be extended to include thermal effects, mechanical deformation, and phase-change phenomena to capture a broader range of aging and failure mechanisms. Integration with experimental microstructure data from tomography or focused ion beam imaging could further enhance model realism and predictive capability. Moreover, coupling with cloud-based green software infrastructures ensures scalability and sustainability, supporting large-scale parametric explorations and accelerated material screening.

In conclusion, this work establishes a versatile, high-fidelity simulation platform that advances computational electrochemistry for lithium-ion batteries. By enabling detailed pore-scale resolution of coupled transport and electrochemical kinetics within realistic microstructures, it provides critical insights to guide the design of next-generation energy storage materials and devices.

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Annex

Table of Variables

Variable	Units	Description
C	mol m^{-3}	Lithium-ion concentration (solid or electrolyte phase)
D	$\text{m}^2 \text{s}^{-1}$	Diffusion coefficient of lithium ions
F	C mol^{-1}	Faraday constant
ϕ	V	Electric potential (solid or electrolyte phase)
U	V	Interfacial potential at the solid–electrolyte interface
I	A m^{-2}	Current density
k	$\text{m}^{5/2} \text{s}^{-1} \text{mol}^{-1/2}$	Reaction rate constant in Butler–Volmer kinetics
α_a	–	Anodic transfer coefficient
α_c	–	Cathodic transfer coefficient
η	V	Overpotential (difference from equilibrium potential)
σ	S cm^{-1}	Electrical conductivity (solid phase)
κ	S cm^{-1}	Ionic conductivity (electrolyte)
R	$\text{J mol}^{-1} \text{K}^{-1}$	Universal gas constant
T	K	Temperature
t	s	Time
x, y, z	m	Spatial coordinates in 3D domain
τ	–	LBM relaxation time parameter
c_i	–	Discrete velocity direction in the LBM lattice
Δx	m	Spatial grid step in LBM model
Δt	s	Time step in LBM model
C_e	mol m^{-3}	Electrolyte lithium-ion concentration
C_s	mol m^{-3}	Solid-phase lithium concentration
C_{max}	mol m^{-3}	Maximum lithium concentration in solid phase
C_{surf}	mol m^{-3}	Surface lithium concentration